Response to Reviewers

Manuscript: *Proxy-Based Diagnostics of Quantum Oracle Sketching Robustness for Non-IID Sensor and Telemetry Streams* (submitted 16 July 2026 as *Quantum Oracle Sketching for Non-IID Sensor and Telemetry Streams: A Proxy-Based Computational Study*) **Journal:** Sensors (MDPI) · **Manuscript ID:** [ID] **Date:** 11 August 2026

We thank both reviewers for careful, constructive reviews that have materially improved the paper. Both reviews converge on the same central point, and we agree with it: the previous version, although internally hedged, still presented itself as evidence about quantum performance, when what it can honestly support is a **proxy-based diagnostic and hypothesis-generation framework for correlation-sensitive streaming analysis**. The revision repositions the manuscript accordingly — title, abstract, statistics, figures, and conclusion — without removing any experiment or altering any measured result: all stochastic outputs regenerate bit-identically from the same seeded pipeline (we verified this explicitly before re-analysis; the verification scripts ship with the revision).

Summary of the principal changes (full detail under each point below):

1. **Retitled** to *Proxy-Based Diagnostics of Quantum Oracle Sketching Robustness for Non-IID Sensor and Telemetry Streams*; the abstract now opens "In theory, …", states that no quantum algorithm is implemented or simulated, and was shortened to ~220 words. *(Note on the tracked-changes file: the MDPI class places the title and abstract in the document preamble, which latexdiff does not mark; both are completely rewritten.)*
2. **New Section 3.3, "Relationship Between Operational Proxies and Quantum Oracle-Sketching Theory"**, separating established theory / assumptions / heuristic constructions / scope limitations, with a provenance table (Table 1) stating per quantity what is inherited, what is introduced, and what is *not* theoretically guaranteed — including the explicit statement that under the framework's own r-only accounting no proxy–baseline crossover exists, so the τ-penalised landscape is a classical effective-sample stress test, not the framework's cost model.
3. **Statistics rebuilt around estimation rather than testing** (Experiment 5): Hodges–Lehmann paired-difference estimates with exact signed-rank 95% intervals, matched-pairs rank-biserial correlations (interpreted as sign-concordance), p-values demoted to descriptive companions, disclosure that no comparison survives Holm correction at $n = 10$ (with the reason), the τ-estimator lattice sensitivity, and an explicit statement that the crossover location is a deterministic solution whose uncertainty comes from the τ estimator and the constants, not from tests.
4. **New sensitivity analyses from existing data** (no new stochastic runs): a C–δ grid for the crossover (Table 8: rho* moves from 0.38 to beyond the sweep; ordinal reading is the robust one) and an encoder/resolution sensitivity table for the real streams (Table: 10 re-encodings; the NYC-above-Machine-Temperature ordering holds in all 10).
5. **New Discussion subsections** on why analytic and empirical correlation estimates diverge, and on what the landscape can and cannot predict — including three **falsifiable hypotheses (H1–H3) with explicit refutation criteria**.
6. **New "Threats to Validity" section** (construct / internal / external / statistical-conclusion) and a promoted, expanded **Future Work** section.
7. **Conclusion rewritten**: primary contribution = the (τ, r) landscape; secondary = proxy-based hypothesis generation; the sentence "The paper demonstrates no quantum advantage and implements no quantum algorithm" appears verbatim.
8. **Figures**: the Experiment-5 legend entry "Proxy-model advantage" renamed "Proxy-estimated separation"; in-image labels "n = 10 qubits" / "Quantum O(n)" removed from Figures 1, 2, 6; captions

aligned with the proxy framing; Figure 8(d)'s synthetic markers disclosed as schematic; typographic and overfull fixes throughout.

9. **Reference audit (preprints)**: all 54 references were classified and every preprint-linked entry checked for a peer-reviewed version. One citation was upgraded — Kallaugher, Parekh and Voronova (SOSA 2025) now carries the SIAM proceedings DOI (10.1137/1.9781611978315.2) instead of its arXiv link. One preprint is retained and explicitly flagged: Zhao et al. (arXiv:2604.07639) is the framework this paper studies; we verified that no peer-reviewed version exists to date, the bibliography entry is labelled "Preprint", and every invocation of its theorems is co-cited with peer-reviewed anchors (Kallaugher, FOCS 2021; Huang et al., Science 2022; Kallaugher et al., SOSA 2025). A citation–bibliography consistency check confirms all 54 in-text keys map one-to-one onto the 54 entries.

A tracked-changes PDF (`main_tracked_changes.pdf`, latexdiff against the reviewed version) accompanies this letter.

Reviewer 1

We thank the reviewer for the structured list of fifteen recommendations; every one is addressed below.

R1.1 — "I strongly recommend to reduce the emphasis on quantum advantage and instead presenting the τ,r landscape primarily as a diagnostic framework to identify scenarios where future quantum implementations is most promising."

Response. Adopted as the organising principle of the revision. The title now leads with "Proxy-Based Diagnostics"; the Introduction states the paper "is best read as a diagnostic and hypothesis-generation framework … it neither demonstrates nor claims a realised quantum–classical separation" (§1); the Conclusion names the landscape as the primary contribution and hypothesis generation as the secondary one; and §7.4 now states three falsifiable hypotheses (H1–H3) that make "where future implementations are most promising" concrete and testable. *Changes:* Title; Abstract; §1 (scope paragraph, contributions 3–4); §7.4 (Scope of the Diagnostic, incl. H1–H3); §10 (Conclusions, rewritten); every "quantum-favourable" replaced by "proxy-favourable"/"hypothesized favourable".

R1.2 — "Please clarify the relationship between the heuristic proxy model and the original theoretical framework."

Response. A dedicated subsection now does exactly this. §3.3 separates (i) *established theory* — the ×R sample overhead, O(n)-qubit memory, and task-specific O(N^c) classical hardness, all theorems of the original framework; (ii) *assumptions* this paper introduces; (iii) *heuristic constructions* — the additional τ division in T_eff (deliberately more pessimistic than the framework's own accounting, under which no crossover would exist at all) and the VC-type task rate used in place of oracle-construction cost; and (iv) *scope limitations*. Table 1 tabulates the provenance and epistemic status of every model-level quantity, with a "not guaranteed" column. *Changes:* New §3.3 with four labelled paragraphs and Table 1; §3.2 proxy-status paragraph.

R1.3 — "The analytic Markov τ proxy differs substantially from the empirical measurements. I recommend to discuss this discrepancy in greater detail and explain whether it reflects limitations of the proxy model or the synthetic data generation process."

Response. A dedicated subsection (§7.3, "Why Analytic and Empirical Correlation Estimates Diverge") now answers the question directly. For Markov switching the two quantities answer different questions: each *bit's* marginal autocorrelation decays as ρ^k and crosses 1/e at ≈2.8 lags (ρ = 0.7), while the *joint* n-bit chain takes Θ(n log n/(1−ρ)) steps to mix — both numbers are correct answers to different questions. The divergence is therefore partly expected behaviour and partly a limitation of single-scalar proxies; it is **not** a data-generation defect: the generators reproduce their intended statistics, and the gap persists under all four alternative τ estimators of Experiment 9, isolating the estimator family as the cause. The subsection also derives the practical implications (empirical τ as the operational input; analytic τ as a conservative envelope; their ratio as a long-memory diagnostic). *Changes:* New §7.3 (three labelled paragraphs: technical explanation / interpretation / implications); Experiment 4 cross-reference; Conclusion RQ1.

R1.4 — "I recommend to validate the proposed framework on additional real-world datasets. It will help to better demonstrate its generalizability."

Response. We agree this is the most valuable extension, and we are explicit that we have not done it within this revision: adding datasets would have been a new experimental campaign, and we preferred to keep this revision strictly re-analysis-based so that every published number remains bit-reproducible. What we did add, from the same raw data, is an **encoder/resolution sensitivity analysis** (new table in §6.8): the two streams were re-encoded across ten sampling-resolution and bucket-width configurations, and the NYC-Taxi-above-Machine-Temperature proxy ordering holds in all ten — so the existing two-stream placement is at least not an artefact of one encoder setting. The manuscript now states plainly that the real-stream experiment "is illustrative rather than statistically representative" (§6.8), the Threats section records the $n = 2$ limitation, and Future Work item (iii) commits to network telemetry, cybersecurity event streams, financial series, and IoT fleets, with hypothesis H1's protocol (rank correlation between placement and outcome over a $\geq$6-stream corpus) defining exactly how generalisability will be tested. *Changes:* §6.8 (illustrative statement; encoder/resolution table and paragraph); §8 External validity; §9 Future Work (iii); §7.4 H1–H2.

R1.5 — "Please include a more comprehensive sensitivity analysis of the selected proxy parameters, particularly C and δ, by this you will show how they influence the reported results."

Response. Done, quantitatively. Because the proxy is a closed-form function of the measured (τ, r), we re-evaluated it on the same per-seed empirical τ values across C in $\{1, 1.5, 2, 2.5\}$ crossed with δ in $\{0.01, 0.05, 0.1\}$. New Table 8 reports the induced movement of the proxy–baseline crossover: regime *ordering* is invariant (C and δ enter only through the monotone factor $\kappa = C^2\log(1/\delta)$), but the crossover location moves from $\rho^* \approx 0.38$ at $(C = 2.5, \delta = 0.01)$ to beyond the sweep range at $C = 1$, with $\rho^* \approx 0.82$ at the operating point. The text now instructs that the landscape be read **ordinally**, the abstract carries the conditionality explicitly, and we additionally state that $C = 2$ is a conventional order-unity choice not fixed by any constant in the cited theory (§5.3). *Changes:* New "Proxy constants C and δ" paragraph + Table 8 in §6.9; §3.2 and §5.3 updated; Abstract; §8 Statistical-conclusion validity.

R1.6 — "The comparison between deterministic quantum proxies and stochastic classical baselines should be presented more cautiously. The current statistical analysis may unintentionally suggest a comparison between implemented algorithms rather than between empirical results and theoretical proxies."

Response. We rebuilt the statistical presentation around this point. Table 4 now reports Hodges–Lehmann estimates of the paired (proxy − classical) difference with **exact** signed-rank 95% intervals and rank-biserial correlations; p-values are retained only as descriptive companions. A dedicated paragraph ("What these tests do and do not establish", §6.5) states that the pairing is stochastic-learner-versus-model-prediction, that the proxy's seed variability comes solely from the per-seed τ estimate, that r_{rb} here indexes *sign concordance across seeds* rather than magnitude, and — verbatim — that the tests "do not, and cannot, establish that one algorithm outperforms another, and no quantum algorithm is implemented anywhere in this comparison". We further disclose that none of the five comparisons survives Holm correction at $n = 10$ (smallest adjusted $p = 0.098$; an outcome of one-to-two discordant seeds, not a design ceiling), that the $\rho = 0.48$ interval is not robust to a single step of the integer-valued τ estimator, and that the crossover location is the deterministic solution of acc_proxy($\hat{\tau}(\rho)$) = mean classical accuracy, with uncertainty inherited from $\hat{\tau}$ and (C, δ), not from tests. *Changes:* Table 4 (rebuilt, incl. caption stating the five pre-specified test points); §6.5 body and interpretation paragraph; Abstract (p-value removed); §7.1; §8 Statistical-conclusion validity.

R1.7 — "I recommend to provide a stronger justification for the selected classical baselines and discussing if additional streaming methods can offer a more informative comparison."

Response. §5.2 now justifies the trio explicitly: the three baselines span complementary axes of bounded-memory streaming learning — gradient-based (online SGD), variance-reduced gradient-based (averaged SGD), and non-gradient hash-count-based (Count-Min) — while sharing the single-pass, fixed-memory regime that the proxy models. We also explain why drift-adaptive methods (ADWIN-equipped learners, adaptive random forests) are deliberately excluded from the primary comparison — they respond to non-stationarity by discarding history, which would confound the correlation-budget question the proxy isolates — and we name benchmarking them on the same landscape as an extension. *Changes:* §5.2 (new justification passage with citations); §9 Future Work.

R1.8 — "Please explain more clearly why the proposed operational proxies, τ and r, are sufficient to characterize the correlation properties of the studied data streams, and discuss any limitations of relying only on these two measures."

Response. The revision treats two-scalar sufficiency as an explicit *assumption*, not a fact: §3.3 ("Assumptions", item (a)) states it as a modelling choice of this paper, and §7.3 shows its precise failure mode — for non-mixing (long-range-dependent) processes no single scalar exists that the analytic and empirical estimators could agree on, so a single-number τ is intrinsically lossy there. The mitigations are stated: the analytic-to-empirical τ *ratio* is itself a useful third diagnostic flagging long-memory content, and multi-scale refinements of τ (integrated-autocorrelation estimates at several horizons) are named as follow-on work. The Threats section (Construct validity) records the residual risk. *Changes:* §3.3 Assumptions; §7.3 Interpretation and Implications; §8 Construct validity; §9 Future Work (ii).

R1.9 — "I recommend to expand the discussion of the assumptions behind each synthetic correlation regime and to please explain how closely these settings represent practical sensor and telemetry environments."

Response. §4 already carried per-regime provenance notes; the Summary (§4.6) now adds the realism mapping: Markov switching idealises short-range state persistence in sensor readouts; seasonal drift the diurnal/weekly cycles of telemetry and demand series; burst repetition the duplicate-heavy event bursts of monitoring and logging pipelines; and long-range dependence the self-similar traffic behaviour first measured on Ethernet — with the explicit caveat that these are idealisations, that real streams mix the mechanisms (as the two NAB cases show), and that the mapping to any concrete deployment is qualitative (Table 9). *Changes:* §4.6 (new passage with citations); Table 9 caption (analytic-convention statement).

R1.10 — "The manuscript would benefit from a dedicated discussion of threats to validity, include the assumptions introduced by the heuristic proxy model, synthetic data generation and feature binarization."

Response. Added as a dedicated section. §8 ("Threats to Validity") is organised by construct, internal, external, and statistical-conclusion validity and covers exactly the requested items: the heuristic proxy constructs (with pointers to the provenance table), synthetic-generator selection, feature binarisation (arcsine shrinkage makes real-stream (τ, r) estimates lower bounds on the continuous-process correlation), the deterministic-vs-stochastic comparison design, dataset selection, target definition (including the look-ahead point raised by Reviewer 2), fixed hyperparameters, and the $n = 10$ seed budget. *Changes:* New §8 replacing the former limitations list.

R1.11 — "I recommend to discuss the computational complexity of estimating τ and r and the scalability of the proposed framework for high-volume streaming systems."

Response. Added. §5.1 now contains an "Estimator cost and scalability" paragraph: the τ estimator needs the bitwise autocorrelation up to lag K — O(nTK) time naively, O(nK) memory with a ring buffer per feature — and the r estimator scans a W-length forward window — O(TW) comparisons, expected O(T) with hashing, O(Wn) memory. Both are linear in T, independent of the alphabet size N = 2^n, single-pass friendly, and embarrassingly parallel across features; on the longest stream studied (T = 22,695) they complete in well under a second, so the diagnostic adds negligible overhead for high-volume deployments. *Changes:* §5.1 (new paragraph).

R1.12 — "Some figures contain many overlapping curves and annotations, I recommend to simplify the visual presentation and enlarge labels to improve readability."

Response. All nine figures were re-audited. Concretely: the Figure 5 legend entry "Proxy-model advantage" was renamed "Proxy-estimated separation" and stray typographic backslashes were removed from five panel titles (Figures 4–6 regenerated); in-image labels "n = 10 qubits", "Quantum proxy (IID, 10 qubits)", and "Quantum O(n)" were replaced with neutral proxy/cited-theory labels (Figures 1, 2, 6 regenerated); Figure 8(d)'s synthetic markers are now labelled as schematic orientation marks and the caption gives the numeric stream coordinates instead of a visual-adjacency claim; caption CI statements were scoped to the stochastic panels only; and four tables that previously overran the text width were reformatted. All regenerations are rendering-only — the numerical outputs are byte-identical. (The landscape figures already use colour-blind-safe palettes with direct contour labelling from the previous revision cycle.) *Changes:* Figures 1, 2, 4, 5, 6 regenerated; captions of Figures 1, 3, 5, 6, 7, 8, 9; Tables 2–9 formatting.

R1.13 — "The discussion could better distinguish theoretical quantum memory advantages from the practical observations obtained through the proxy-based evaluation."

Response. This distinction is now enforced at every occurrence. Experiment 6 opens: "No memory lower bound is measured empirically in this experiment", and states that no result constitutes an empirical memory floor for online SGD, averaged SGD, or Count-Min; §3.3 ("Memory references") makes the same statement structurally; the provenance table lists the $\Omega(\sqrt{N})$ curve as a "conservative theoretical reference" whose empirical-floor status is explicitly not guaranteed; Figure 6's caption is labelled "cited bounds; no measured content"; and §7.1 closes its three structural factors with "All three factors are properties of the proxy model and the cited theory; none is a measured quantum result", with the $n \geq 20$ statement marked as an extrapolation beyond the tested range. *Changes:* §3.3; §6.6 (opening and panel text); Figure 6 caption; §7.1; Table 1 row.

R1.14 — "Please discuss the applicability of the proposed framework to other streaming domains beyond sensor telemetry, such as network traffic, financial streams, or cybersecurity monitoring."

Response. Table 9 maps network backbone traffic, financial returns, sensor/IoT, and concept-drifted streams onto the landscape from published measurement ranges (now recomputed from the paper's own formula, with verdicts given as ranges and the estimator convention stated), and Future Work item (iii) commits to exactly the requested domains: network telemetry (flow and latency series), cybersecurity event streams (intrusion and log-anomaly feeds), financial tick and returns series, and large-scale IoT monitoring fleets. *Changes:* Table 9 (recomputed labels and caption); §7.2; §9 Future Work (iii).

R1.15 — "strengthen the conclusion, for this emphasize the practical contributions of the study, clearly summarize its limitations, and please outline the most important directions for future research."

Response. The Conclusion is rewritten to exactly this structure: a "Practical contribution" paragraph (the landscape as a cheap pre-screening step — estimate (τ, r), place the stream, read off whether correlation alone erodes the budget below the rare-event-learning level, as it does for Machine Temperature at T_eff = 52); a "Limitations and required validation" paragraph pointing to the Threats section and naming the decisive missing step (an implemented or simulated oracle-sketching pipeline evaluated at predicted-favourable and predicted-unfavourable coordinates); and research-question answers restated in effect-size terms with estimator conventions bound. *Changes:* §10 (full rewrite); §9 Future Work.

Reviewer 2

We thank the reviewer for an unusually thorough review; the closing recommendation — make the (τ, r) landscape-as-hypothesis-generator the paper's main message — is precisely the repositioning this revision performs.

R2.1 — "The key issue is the relationship between τ and r and the quantities in the framework of quantum oracle-sketching. … They have to strengthen the theoretical basis for the chosen proxy construction and distinguish the results from the original from the assumptions that are used in this paper."

Response. New §3.3 does this separation explicitly, in four labelled parts: *Established theory* (the ×R overhead theorem, O(n)-qubit memory, task-specific O(N^c) hardness — all results of the original framework, taken as given); *Assumptions* (two-scalar summarisability, binarised-stream fidelity, task-rate-as-pipeline-stand-in — all modelling choices of this paper, none inherited); *Heuristic constructions* (the τ division absent from the framework's accounting; the VC-type rate in place of oracle-construction cost); and *Scope limitations*. Table 1 gives per-quantity provenance with an explicit "not guaranteed" column. We also now state the sharpest consequence: under the framework's own r-only accounting, the Markov, seasonal, and long-memory regimes (r = 1) incur zero overhead and no proxy–baseline crossover exists — the τ-penalised landscape is a classical effective-sample-size stress test, strictly more pessimistic than the framework's accounting, not a rendering of it. *Changes:* New §3.3 + Table 1; §3.2; effective-sample-size lineage now cited (Geyer 1992).

R2.2 — "The terminology of quantum performance should be moderated throughout the paper. … we will refer to them as proxy-based predictions instead of the actual quantum performance. This is particularly relevant in the Abstract, Results, Discussion, figure captions and Conclusion."

Response. A systematic terminology pass was made over exactly those surfaces. "Quantum-favourable" no longer appears (now "proxy-favourable"/"hypothesized favourable"); "quantum performance proxy crosses below classical performance" and similar constructions were rewritten ("the accuracy proxy computed with the analytic τ lies below the measured classical baseline"); Table 6's column is "Proxy hypothesis"; and — because captions alone cannot fix text baked into images — Figures 1, 2, and 6 were regenerated to remove the in-image labels "n = 10 qubits", "Quantum proxy (IID, 10 qubits)", and "Quantum O(n)". Explicit "no quantum algorithm is implemented" statements now appear in the Abstract, §3.3, §6.5, §6.6, §6.8, §7.4, §8, and the Conclusion. *Changes:* Global; Figures 1, 2, 6 regenerated; all captions audited.

R2.3 — "The statistical comparison between classical baselines and the proxy model is still not resolved. Our Wilcoxon tests compare stochastic classical results for seeds to a proxy value that is deterministic for a given value of parameters. … We believe the magnitude of the classical minus proxy difference and the uncertainty may be a better gauge than simply p-values."

Response. Adopted in full — magnitude and uncertainty now lead, and the tests are re-scoped to the one thing they can support. Table 4 reports, per sweep point, the Hodges–Lehmann estimate of the paired difference with its exact signed-rank 95% interval (e.g., +0.042 [+0.011, +0.087] at ρ = 0; −0.046 [−0.079, −0.003] at

$\rho = 0.88$) and the rank-biserial correlation, which we interpret explicitly as sign-concordance across seeds (scale-free against a near-constant reference; all magnitude information lives in the difference estimate). The p-values are descriptive companions only; we disclose that none survives Holm correction at $n = 10$ and why; we disclose the integer-τ lattice sensitivity (one lag step = 0.012, enough to move the $\rho = 0.48$ interval across zero); and the interpretation paragraph states that the crossover is the deterministic solution of $\text{acc\_proxy}(\hat{\tau}(\rho))$ = mean classical accuracy, with uncertainty inherited from the estimator and the constants (C, δ), not from the tests. The full re-analysis (which first reproduced the published per-seed results bit-exactly — all five p-values to six digits) ships with the revision. *Changes:* Table 4 + caption; §6.5 (rewritten body and interpretation paragraphs); Abstract; §7.1; §8; `revision/exp5_reanalysis.py` + output.

R2.4 — "The huge discrepancy between the analytic and empirical refreshing time is one of the most important findings in the paper, especially for the long-range dependence regime. … The authors should discuss this in more detail and discuss its implications for the interpretation of the central (τ,r) sensitivity landscape."

Response. We agree it is one of the paper's most important findings and have promoted it accordingly. New §7.3 gives the mechanism (the analytic proxies target model-level worst-case scales — joint-chain mixing; tail-dominated integrated autocorrelation — while the empirical estimator measures the short-lag marginal decorrelation of the binarised stream; for a non-mixing process there is no scalar the two could agree on), and the implications for the landscape are now carried through the whole paper: verdicts are bound to their estimator convention everywhere, and we state openly that the burst and long-memory extremes *exchange places* between the analytic and empirical conventions (burst saturates through its measured duplication; long-memory's high empirical value is the short-lag artefact) — in Experiment 9, §7.3, the RQ2/RQ3 answers, and the Figure 3/7 captions. The abstract now says the two estimates "diverge" (not "overestimate", which presupposed the empirical value as ground truth) and adds that they answer different questions. *Changes:* New §7.3; §6.9; §6.3; Figures 3, 7 captions; Conclusion RQ1–RQ3; Abstract.

R2.5 — "The classical-memory comparison needs to be taken with care. The classical lower bound curve … is not actually an empirical lower bound for the SGD, averaged-SGD, or Count-Min tools used in our experiments. … we should make it more prominent in the dimension-scaling discussion."

Response. Made prominent exactly there. Experiment 6 now opens with: "No memory lower bound is measured empirically in this experiment: the curves in panel (a) plot cited theoretical quantities, and no result here constitutes an empirical memory floor for online SGD, averaged SGD, or Count-Min." Figure 6's caption labels panel (a) "cited bounds; no measured content" and the in-image curve labels now read "(cited theory)". §3.3's "Memory references" paragraph and the provenance table carry the same statement structurally, and §7.1 states that none of its structural factors is a measured quantum result. *Changes:* §6.6; Figure 6 (caption and regenerated labels); §3.3; Table 1; §7.1.

R2.6 — "The real-stream analysis is useful as a sanity check but it is limited to two NAB datasets. The conclusions on the sensor and telemetry applications should therefore be conservative."

Response. The conclusions are now explicitly conservative: §6.8 states the experiment "is illustrative rather than statistically representative", the External-validity paragraph records that generalisation is unverified, and the abstract claims placement only. Additionally, a new encoder/resolution sensitivity table strengthens what *can* be said from two streams: across ten re-encodings of the same raw series (three sampling resolutions × three bucket widths), the NYC-above-Machine-Temperature ordering holds in every configuration, so the reported placement is at least encoder-stable. Broader corpora are committed in Future Work with a concrete test protocol (H1). *Changes:* §6.8; new encoder-sensitivity paragraph + table; §8; §9; §7.4 H1.

R2.7 — "the target defined based on a percentile of the full time series should also be reconsidered in

the context of streaming analysis, since using the full series to set the target threshold can take into account future observations. A threshold set from a first calibration interval or updated only from previously observed samples would provide a more rigorous prequential design.”

Response. This is an excellent catch, and we now disclose it explicitly rather than leave it implicit. §6.8 (“Binarisation choice”) states that the threshold is computed from the full series and therefore uses information not available prequentially at each step; that the feature stream and the (τ, r) estimates involve no such look-ahead — so the landscape placement, the paper’s central output for these streams, is unaffected, the leakage touching only the classical-baseline reference numbers; and that a threshold calibrated on an initial interval, or updated from previously observed samples only, is the rigorous prequential design, which we adopt as future work. The Threats section (Internal validity, item (v)) records it as a residual risk. We chose disclosure over re-running the baselines in this revision to keep every published number bit-reproducible; we are happy to re-run with a calibration-interval threshold in a further round if the reviewer prefers. *Changes:* §6.8 Binarisation paragraph; §8 Internal validity (v); §9.

R2.8 — “The interpretation of the NYC Taxi Count-Min result also needs to be qualified. The six-bit representation gives a relatively small number of possible feature vectors and hence memorising the same feature patterns is relatively easy. … I would like to give more weight to the implications when we think about the apparent agreement between the Count-Min performance and our proxy prediction.”

Response. The qualification now carries real weight in three places. The dedicated “Why Count-Min looks strong” paragraph is retained; the interpretation adds, plainly: “the favourable-side separation between the two streams rests on Count-Min alone; the hashed-linear learners do not corroborate it” — and we additionally report that both SGD variants fall *below* the majority baseline on accuracy and that Averaged-SGD’s balanced accuracy straddles chance, so the apparent proxy–Count-Min agreement is explicitly not read as validation. §7.4 lists representation-specific effects like this memorisation route among the things the landscape *cannot* anticipate, and — going further in the reviewer’s direction — the same section now concedes that, read strictly as an accuracy projection, the proxy is falsified on Machine Temperature (0.500 predicted vs 0.72–0.80 observed), with the rare-class diagnostic reading offered as interpretation, not confirmation. Table 6 now carries the ordering-to-ordering intervals (Count-Min F1 = 0.644 ± 0.062, per-ordering range 0.43–0.72), which tempers the headline number. *Changes:* §6.8 (findings bullets, Count-Min paragraph, interpretation (ii)–(iii)); Table 6 (intervals restored); §7.4.

R2.9 — “In particular, the target function ablation should be highlighted because it shows that our classical performance can change dramatically with the classification task in question and that our proxy is in fact target independent. A correlation-based proxy that is insensitive to the target function cannot necessarily predict the performance of algorithms for different kinds of decision problems.”

Response. Highlighted and stated as a scope boundary in the reviewer’s own terms. §6.9 (Target function) now reads: “classical learner performance is strongly target-dependent, while the proxy depends only on (τ, r, T) and is target-independent by construction. The proxy therefore cannot universally predict downstream classifier performance; it characterises the correlation structure of the stream — the effective sample budget available to any bounded-memory learner — not task-specific learnability.” §7.4 elevates this to the “cannot” list (with the further empirical point that measured classical accuracy is flat while T_eff falls fivefold, so the budget model does not predict even the classical arm’s accuracy), the Figure 9 caption states the proxy “does not track task-specific learnability”, and RQ2’s answer now says τ and r are budget summaries, not accuracy predictors. The ablation’s protocol (prequential, lr = 0.01, T = 4096) is also now stated so its absolute levels are not misread against the main experiments. *Changes:* §6.9; §7.4; Figure 9 caption; Conclusion RQ2.

R2.10 — “the strongest contribution of the paper is the (τ,r) sensitivity landscape proposed as a hypothesis-generating tool … The paper would be better off having this methodological contribution

as the main message rather than the separation of quantum-classical performance."

Response. This is now the paper's stated identity, end to end: the title leads with "Proxy-Based Diagnostics"; the abstract announces "a proxy-based diagnostic and hypothesis-generation framework"; the contribution list names the landscape "the paper's central diagnostic and hypothesis-generation artifact"; §7.4 states the falsifiable hypotheses it generates; and the Conclusion opens by declaring the landscape the primary contribution and hypothesis generation the secondary one, followed by: "The paper demonstrates no quantum advantage and implements no quantum algorithm." *Changes:* Title; Abstract; §1; §7.4; §10.

Note on verification

Before making any statistical change we reproduced the published Experiment-5 per-seed results bit-exactly from the pinned seeds (all five Wilcoxon p-values match the published values to six significant digits), and all figure regenerations are rendering-only (numerical outputs byte-identical). The re-analysis and sensitivity scripts, with their outputs, are included in the revision package (`revision/exp5_reanalysis.py`, `revision/encoder_sensitivity.py`, and `code/exp5_effectsizes.py` in the repository), and the reproducibility documentation was corrected and extended (exact seed indices per experiment; the previously missing Machine-Temperature download step).

On behalf of all authors,

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